KARAN KUMAR

Hyderabad, Telangana | 9154994175 | karankumar21143@gmail.com
Portfolio | GitHub | LinkedIn | LeetCode

WORK EXPERIENCE

Frontend Developer Intern | Galactix Solutions Pvt. Ltd. **Mar 2026 – Jun 2026**

- Developed 20+ reusable React.js components and integrated 15+ backend REST APIs using Redux Toolkit, managing API calls, application state, asynchronous data, and frontend–backend data synchronization.
- Wrote 50+ Jest unit tests for React components and API-driven workflows, increasing test coverage from 45% to 80% and improving regression detection during feature development.
- Resolved 100+ SonarQube code-quality issues, including refactoring functions flagged for cognitive complexity above 18, improving maintainability and reducing technical debt.
- Collaborated with back-end developers in Agile sprints (planning, code reviews, standups) to deliver frontend features and resolve production issues.

Research Intern | IIIT Hyderabad (IIITH) **Jun 2026 – Oct 2026**

- Built and evaluated a cascaded Telugu speech-to-speech translation pipeline (ASR → MT → TTS) on the IISc RESPIN corpus using Vanni FastConformer, IndicTrans2, and Spring_F5 for a direct S2ST research project.
- Benchmarked ASR over 775K+ utterances (~1,150 hours) with Python and JiWER, and fine-tuned Vanni FastConformer in PyTorch/Hugging Face on Banking and Agriculture speech (49.5% in-domain vs 64.4% cross-domain WER).
- Quality-scored ~930K Telugu→English/Hindi translations (CometKiwi, COMET-QE) and synthesized speech with Spring_F5; built reproducible, traceable evaluation workflows under limited GPU compute.

PROJECTS

ShopEase - Full-Stack E-Commerce Website | React.js, Redux Toolkit, Tailwind CSS, Node.js, Express.js, MongoDB

- Built a responsive MERN-stack e-commerce application with product listing, search, category and price filters, product detail pages, shopping cart, wishlist, and a multi-step checkout flow.
- Managed cart, user, and product state with Redux Toolkit; consumed REST APIs and JSON data with loading, error, and empty-state handling for a smooth user experience.
- Implemented JWT-based sign-up/login, protected routes, and an admin panel to add, edit, and delete products, with a mobile-first layout built using Tailwind CSS.
- Optimized performance with code splitting, lazy-loaded routes and images, and debounced search.

MovieScope - Movie Discovery App | React.js, JavaScript (ES6+), Tailwind CSS, REST API

- Developed a responsive movie explorer that consumes a public REST API to show trending, top-rated, and searchable movies with genre filters, detail pages, and infinite scroll.
- Built reusable components and custom hooks for data fetching, debounced search, and watchlist storage in localStorage; deployed on Vercel.

TaskBoard - Kanban Task Manager | React.js, Redux Toolkit, Tailwind CSS

- Created a Kanban-style task management dashboard with drag-and-drop columns, task create/edit/delete, priority labels, and filters, with state managed in Redux Toolkit.
- Designed a responsive, keyboard-accessible UI with localStorage persistence and a dark mode toggle.

Mini Vercel - Deployment Platform | React, Node.js, Express.js, AWS S3, Docker

- Built a full-stack platform that takes a frontend project from source to a hosted build, with a React interface and REST APIs for project management, deployment workflows, and live status tracking.
- Moved long-running builds off the request path with async workflows; stored build artifacts in AWS S3.

TECHNICAL SKILLS

- **Core Languages:** HTML5, CSS3, JavaScript (ES6+), TypeScript
- **Frameworks & Libraries:** React.js, Next.js, Redux Toolkit
- **Styling & Design:** Responsive Web Design, Tailwind CSS, Flexbox, CSS Grid
- **APIs & Data:** RESTful APIs, JSON, Async/Await, WebSockets, Socket.IO
- **Tools & Version Control:** Git, GitHub, Browser Developer Tools, Jest, SonarQube, Docker, Linux
- **Back-End Basics:** Node.js, Express.js, JWT Authentication, MongoDB, PostgreSQL, MySQL, Redis
- **Other:** Python, C++, SQL, Data Structures & Algorithms, OOP, System Design

EDUCATION

B.Tech, Computer Science and Engineering **Expected Jun 2027**

Hyderabad Institute of Technology and Management, Hyderabad | CGPA: 8.5/10

Intermediate (12th) | Narayana Junior College, Hyderabad **2021 – 2023**